Tutorial Week 8 - Clustering + PCA

Clustering

1. Which below statement is false regarding Clustering?
   A. A dissimilarity (similarity) function between samples is provided.
   B. groupings of samples into clusters are evaluated by a loss function.
   C. The most likely cluster will be calculated according to hidden states.
   D. There is an algorithm that optimizes this loss function.

2. Which below statement is True regarding Clustering?
   A. Clustering is Unsupervised Learning.
   B. We must label the raw data.
   C. Low-dimensional features in the data might be sufficient to describe the samples.
   D. In the final stage of an investigation, can perform exploratory data analysis to gain some insight into the nature or structure of data.

3. Which of the following metrics do we have for finding dissimilarity between two clusters in "hierarchical clustering"?
   A. Single Linkage
   B. Complete Linkage
   C. Average Linkage
   D. All the above

4. Suppose we have three cluster centroids $\mu_1 = [1, 2]$, $\mu_2 = [-3, 0]$ and $\mu_3 = [4, 2]$. Furthermore, we have a training example $x^i = [-1, 2]$. After a cluster assignment step, what will C^i be?

PCA

1. Principal Component Analysis (PCA) is an example of?
   a) Supervised Learning
   b) Unsupervised Learning
   c) Semi-Supervised Learning

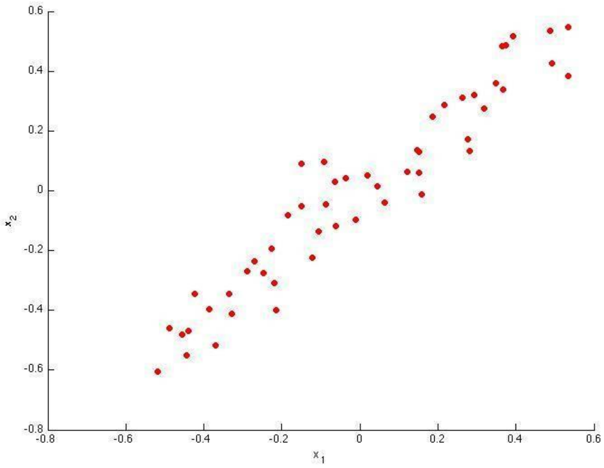
2. What is the importance of using PCA before the clustering? Choose the most complete answer.
   a) Find the explained variance.
   b) Find good features to improve your clustering score.
   c) Avoid bad features.
   d) Find which dimension of data maximize the features variance.

3. When uses PCA?
   a) Everytime before uses a Machine Learning algorithm.
   b) You want to find latent features and reduce dimensionality.
   c) When you have an overfit case.
   d) When the data is small and with a few features.

4. What PCA does Afterall?
   a) Create clusters in order to let you know what are the class.
   b) Predicts your target with high efficiency.
   c) Reduce dimensionality of the data and create new features from features set given.
   d) Give you the highest number of features possible, to maximize the efficience of your Machine Learning algorithm.

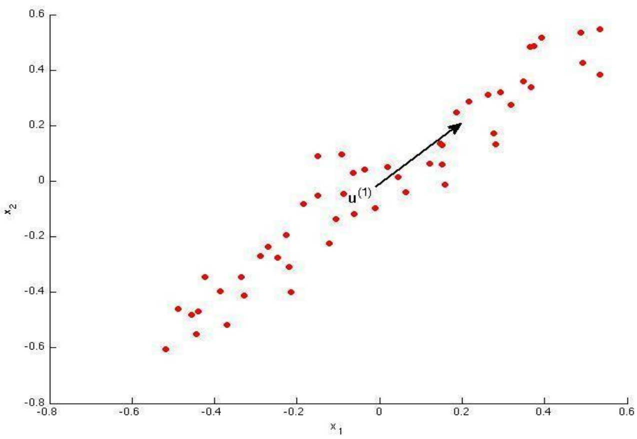
5. Why you have to drop unimportant features?
   A. To trains the model faster.
   B. Find the correct clusters.
   C. Using the most important features will give you better efficiency predicting your target.
   D. Standardize the data.

6. Consider the following 2D dataset:

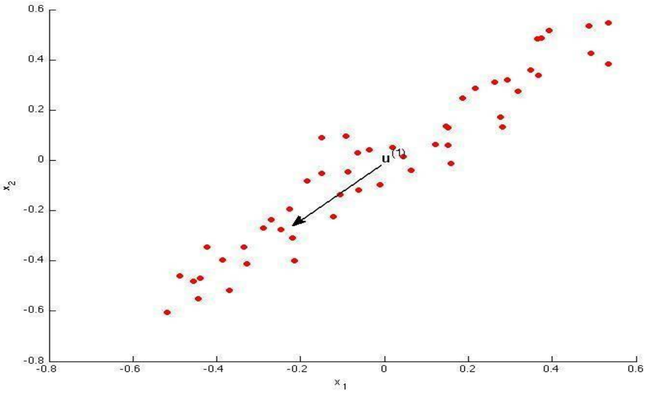


Which of the following figures correspond to possible values that PCA may return for $u^{(1)}$ (the first eigen vector / first principal component)? Check all that apply (you may have to check more than one figure).

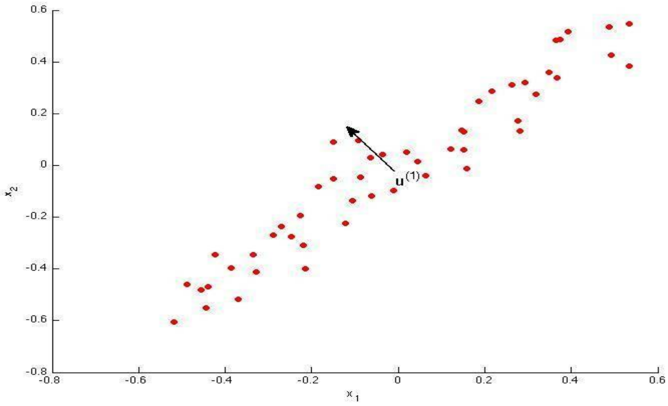
a) Figure 1



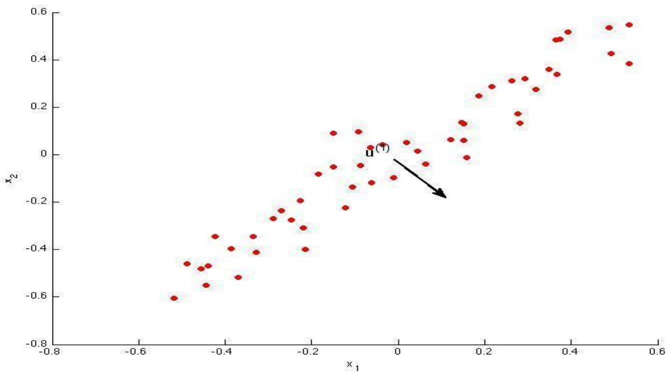
b) Figure 2



c) Figure 3



d) Figure 4



7. Which of the following are recommended applications of PCA? Select all that apply.
   a) To get more features to feed into a learning algorithm.
   b) Data compression: Reduce the dimension of your data, so that it takes up less memory / disk space.
   c) Preventing overfitting: Reduce the number of features (in a supervised learning problem), so that there are fewer parameters to learn.
   d) Data visualization: Reduce data to 2D (or 3D) so that it can be plotted.

e) Data compression: Reduce the dimension of your input data $x^{(i)}$, which will be used in supervised learning algorithm (i.e., use PCA so that your supervised learning algorithm runs faster).
f) As a replacement for (or alternative to) linear regression: For m
g) Data visualization: To take 2D data, and find a different way of plotting it in 2D (using k=2)

8. Which of the following statements are true? Check all that apply.

a) Given only $z^{(i)}$ and U_{reduce}, there is no way to reconstruct any reasonable approximation to $x^{(i)}$.
b) Even if all the input features are on very similar scales, we should still perform mean normalization (so that each feature has zero mean) before running PCA.
c) Given input data $x \in R^n$, it makes sense to run PCA only with values of k that satisfy $k \leq n$. (In particular, running it with $k = n$ is possible but not helpful, and $k > n$ does not make sense.)
d) PCA is susceptible to local optima; trying multiple random initializations may help.
e) PCA can be used only to reduce the dimensionality of data by 1 (such as 3D to 2D, or 2D to 1D).
f) Given an input $x \in R^n$, PCA compresses it to a lower-dimensional vector $z \in R^k$.
g) If the input features are on very different scales, it is a good idea to perform feature scaling before applying PCA.
h) Feature scaling is not useful for PCA, since the eigenvector calculation.

Fill in the blanks in the bellow statements:

9. The first principal direction u_1 must be the ……………… of $\sum \quad X$ That corresponds to the largest value(e_1).

a) eigenvector
b) projection
c) eigenvalue
d) orthogonal

10. PCA linearly combines the variables and let to drop projections that are ……………….

a) latent
b) Less informative
c) Normalized
d) Standardized

11. Doing PCA to avoid overfitting …………..

a) is efficient.
b) is a bad idea.
c) increases accuracy.

Questions 12-16 are related to PCA for Face Recognition.

12. The images are ……………… to line up the eyes, mouths and other features.

a) normalized
b) standardized
c) weighted

13. Eigenfaces are the eigenvectors of the …………… of the vector space of human faces

a) variance
b) variable matrix
c) covariance matrix

14. Eigenfaces are the ‘standardized face ingredients’ derived from the …………….. of many pictures of human faces.

a) statistical analysis
b) feature
c) regularization

15. When properly………….., eigenface scan be summed together to create an approximate grayscale rendering of a human face.

a) standardized
b) normalized
c) weighted

16. Eigenfaces provide a means of applying data …………….. to faces for identification purposes.

a) Compression
b) Visualization
c) Extraction